User query

Describe what you want to create…

Send

- Qubit budget
- Code family
- Weight limit
- Objective
- Connectivity

Orchestrator

Planning | Routing | Memory

Ranked code

- Discovered code
- Compiled circuit
- Decoding performance

FAST LOOP

Seconds

Generative search agent

Construct programs

```
def construct(params):
    n, k = params()
    H_X = sample_HX(n)
    H_Z = sample_HZ(n)
    check_commute(H_X, H_Z)

    Return CSS(H_X, H_Z)
```

- Lifted product
- Bivariate bicycle
- Trivariate tricycle
- Generalized lifted product

Code characterization

$$H_X = \begin{pmatrix} 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}$$

Rank candidates

Cheap surrogate

$$H_Z = \begin{pmatrix} 1 & 1 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 & 1 & 0 & 1 \end{pmatrix}$$

Code-level evaluation

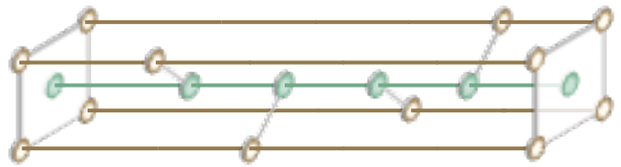
- FOM kd^2/n
- Logical qubits k
- Code distance d

Promote top candidates

SLOW LOOP

Minutes / hours

Circuit synthesis



Circuit evaluation

DEM

err(0.10) D0 D1
err(0.07) D2
err(0.18) D1 D3
err(0.03) D4 L0
err(0.09) D5 D6

Syndrome



Decoders

- BP+OSD
- BP+LSD
- Relay-BP

LER



Feedback for evolving the next generation